Should the Fusion Weight Change with the Question?

The Whole Study: Q1–Q4

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Team and my part

Member	Main contribution	
Jianxiang He	**Problem and setup; score cache and data pipeline; Q1 analysis; overall study (Q1–Q4)**	**this talk**
Longtao Jiang	Continual learning experiments and analysis (Q2–Q3)	
Makhmud Assentay	Downstream question-answering evaluation (Q4)	

Background: two clues, one weight

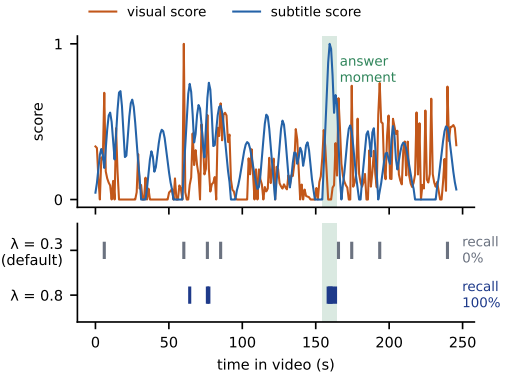
A VLM can only look at **about 8 frames** of a long video, so we must pick them well.

VSI scores every second with two clues:

- **Visual**: do the question's objects appear?
- **Subtitle**: does the subtitle match the question?

$$S(t) = (1 - \lambda)\,\text{visual}(t) + \lambda\,\text{subtitle}(t)$$

and keeps the top 8 seconds. VSI uses $\lambda = 0.3$ **for every question**.



Real example: $\lambda = 0.3$ **misses** the answer moment (green), $\lambda = 0.8$ **hits** it.

Our questions and my part

Four research questions

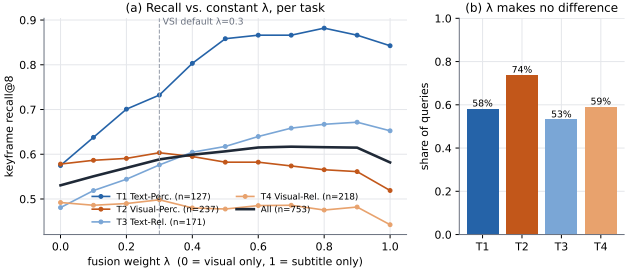
- Q1 Does the best λ depend on the question type?
- Q2 Can a predictor learn λ as question types arrive one by one (continual learning)? Does it forget?
- Q3 If the gain is small, why?
- Q4 Do better frames give better answers?

My part

- Set up the problem and the benchmark: 753 LongVideoBench questions, 4 question types, 3 orders $\times$ 3 seeds
- Built the score cache and data pipeline
- Q1 analysis, and tied Q1–Q4 into one study

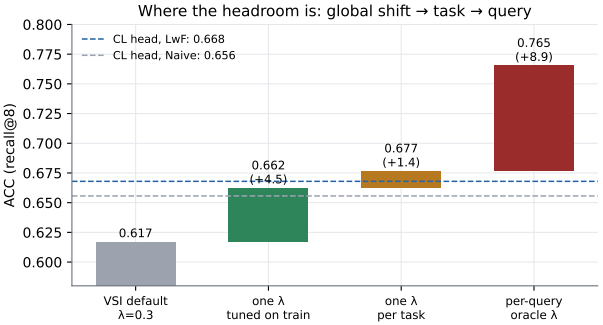
Longtao led Q2–Q3, Makhmud led Q4.

Q1: types want different weights



- Text types best at $\lambda = 0.8$–0.9; visual types at ≤ 0.3
- Per question: text **0.62** vs. visual **0.49**, $p = 1.6 \times 10^{-6}$
- But **53–74%** of questions do not react to λ

Q1: where is the room to improve?

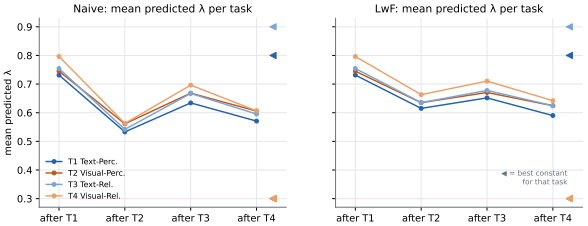


- Tune one global λ: **+4.5**
- One λ per type: **+1.4** more
- Best λ per question: **+8.9** more

Q1 answer

Yes, the type matters; but most of the room is in **per-question** differences.

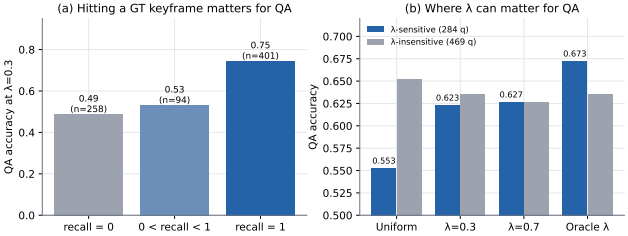
Q2–Q3: ties a tuned constant



Predicted λ per type after each stage; triangles = best λ per type.

- Beats default $\lambda = 0.3$ by **4–5 points** in 9/9 runs
- vs. one **tuned** λ: only **+0.6** (best method)
- Reason: it gives **every type almost the same** λ ($\approx$0.6)
- Forgetting is small; LwF halves it, but only in 6/9 runs

Q4: helps only some questions



- Qwen3-VL-8B: learned $\lambda =$ default ($+0.4$, n.s.)
- Hitting the answer moment matters: **0.49** vs. **0.75**
- On the 284 questions where λ changes the frames, the best λ adds **5 points**

Summary

1. **The best weight depends on the question type**, but most of the gain is per question.
2. **Continual learning beats the default but only ties a tuned constant**; it collapses to one weight for all types.
3. **Better frames help answers only where the weight changes the frames** (about 38% of questions).
4. **Lesson:** always compare a learned weight with a *tuned* constant. Four silent bugs each reversed our conclusions along the way.

Thank You

Questions?

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Backup: setup

Data: LongVideoBench val, one question per video, 753 questions

Types: T1 text perception (127), T2 visual perception (237), T3 text relation (171), T4 visual relation (218)

70/30 train/test split; orders T1-T2-T3-T4, T1-T3-T2-T4, T3-T1-T2-T4; 3 seeds

Predictor: 2-layer MLP, 54 inputs from the two score curves, outputs a weighted average over 11 values of λ

Methods: Naive, EWC, LwF, ER, kNN, Joint (reference)

Baselines: default 0.3; one λ tuned on train; one λ per type; per-question oracle

Metrics: recall@8 (answer moments hit within ± 5 s); ACC = final average; FM = forgetting

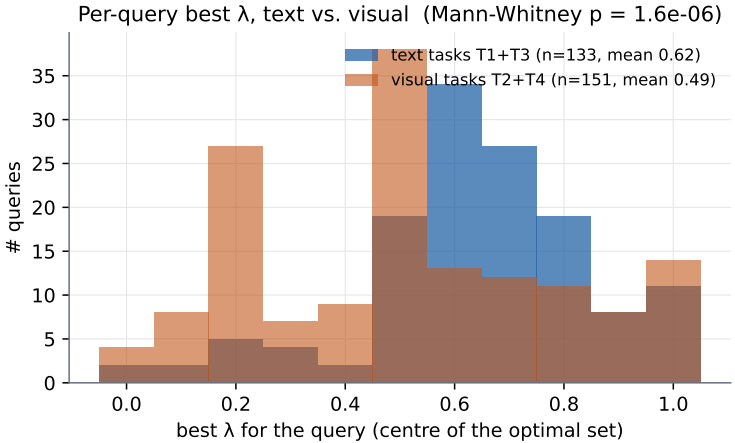
Backup: main results (9 runs)

Method	ACC ↑	FM ↓	vs. 0.3	vs. tuned
Default $\lambda = 0.3$	0.617	–	–	−4.5
One tuned λ	0.662	–	+4.5	–
One λ per type	0.677	–	+6.0	+1.4
Oracle	0.765	–	+14.8	+10.3
Naive	0.656	0.029	+3.9	−0.7 (3/9)
EWC	0.662	0.023	+4.5	−0.1 (4/9)
LwF	0.668	0.015	+5.1	+0.6 (7/9)
ER	0.660	0.021	+4.3	−0.3 (4/9)
Joint (ref.)	0.668	0.015	+5.2	+0.6 (6/9)
kNN	0.645	0.005	+2.9	−1.7 (0/9)

Backup: four silent bugs

Problem	What happened
Frames vs. seconds	Annotations in frames, scores per second: 94% of questions got zero recall.
Annotation frame rate	LongVideoBench counts frames at a fixed **24 fps**, not the real 30 fps (38/39 vs. 14/39 match the answer subtitle).
Short videos dropped	Videos $\leq$16 s could not fill VSI's 4$\times$4 grid and were silently skipped (197 questions).
Train vs. test weight	Trained on the average λ, tested on the most likely one ($\lambda = 0$ or 1): learned weights looked 3–5 points **worse** than the default.

Backup: text vs. visual



Backup: run by run

